

Rich and Lean Representations of Information for Knowledge Work: The Role of Computing Packages in the Work of Classical Scholars

KAREN RUHLEDER
Worcester Polytechnic Institute

“You shall see them on a beautiful quarto page,
where a neat rivulet of text shall meander through a meadow of margin.”

Sheridan, *The School for Scandal* (1777)

Applying information systems to complex intellectual tasks requires the representation and codification of ambiguous and fragmentary forms of data. This application effects changes not only in representation of this data, but in the relationships between users and tools, techniques, or systems for data interpretation. It also affects the complex infrastructures that support this process. This article uses a *package* metaphor to examine the impact on one domain of knowledge work, classical scholarship, of the “computerization” of a key data source, the textual edition. The construction of one on-line textual databank, the *Thesaurus Linguae Graecae* (TLG), has altered the traditional relationships between text “owners” and “users,” has changed the role of the text as a conduit for social and historical information, and has disrupted traditional patterns of transmitting domain expertise. A rich information resource has become lean in its electronic form.

The TLG has standardized the corpus of Greek literature and eased access to a broad range of works, including rare and out-of-print materials. At the same time, its construction has decoupled often-contested textual sources from their accompanying critical notes and supplemental materials. The use of the TLG has also shifted notions of objectivity, accuracy, and requisite expertise within the community. The transmission of domain knowledge must now be coupled with the transmission of technical knowledge, a process for which no infrastructure is currently in place. These experiences parallel those of other knowledge workers “Mechanistic” paradigms of information and knowledge cannot accommodate important components of computing packages, including the transmission of expertise and infrastructures for tool development and evaluation. Recent developments in information storage and dissemination, including gophers and ftp sites, may indicate that despite technical advances that could be used to support rich representations (such as hypertext and multimedia), leaner forms of data may prevail.

Categories and Subject Descriptors: H.1.1 [**Models and Principles**]: Systems and Information Theory; H.1.2 [**Models and Principles**]: User/Machine Systems; H.3.5 [**Information Storage and Retrieval**]: Online Information Services; K.3.m [**Computers and Education**]: Miscellaneous; K.4.3 [**Computers and Society**]: Organizational Impacts; K.6 [**Computing Milieux**]: Management of Computing and Information Systems

Author's address: Department of Management, Worcester Polytechnic Institute, 100 Institute Road, Worcester, MA 01609; email: ruhleder@wpi.wpi.edu.

Permission to copy without fee all or part of this material is granted provided that the copies are not made or distributed for direct commercial advantage, the ACM copyright notice and the title of the publication and its date appear, and notice is given that copying is by permission of the Association for Computing Machinery. To copy otherwise, or to republish, requires a fee and/or specific permission

© 1994 ACM 1046-8188/94/0400-0208\$03.50

ACM Transactions on Information Systems, Vol. 12, No. 2, April 1994, Pages 208–230.

General Terms: Human Factors, Management

Additional Key Words and Phrases: Computerization of knowledge work, computing packages, decontextualization of information, information representations, locus of expertise

1. INTRODUCTION

This article focuses on the impact information systems (IS) may have on the representation and utilization of data by people engaged in complex intellectual tasks. It reports specifically on findings drawn from a study of classical scholarship, where the use of ambiguous and fragmentary textual data forms a central component of work. Applying a *package* metaphor [Kling and Scacchi 1979], the article demonstrates how a shift in the form of a key artifact used by classicists (the “computerization” of the textual edition) disrupts other aspects of the complex *package* that supports its creation, dissemination, and use. These findings are important to the IS community as it tries to codify and utilize similarly ambiguous forms of data for use by other knowledge workers. As more tools are developed to support knowledge work, the IS community must develop a better understanding of the effects of computerization (1) on the codification of data used to accomplish a particular set of tasks, (2) on the relationship between users and the tools, techniques, or systems they use to access and interpret this data, and (3) on the complex infrastructures that support this process.

Classical scholarship is one of many domains of knowledge work currently undergoing an important transition from paper-based tools to computer-based tools. Until recently, paper-based textual editions and concordances formed the primary tools to support work. Now, computerized databanks and search routines redefine the information available to the scholar and alter the relationship between scholar and text. Computer-based tools not only affect the accessibility of various forms of information, but couple or link this information in nontraditional ways, changing classicists’ ways of thinking about the materials they use in their work. Their construction, dissemination, and application also alter the traditional relationships between text “owners” and text “users,” alter the role of the text as a conduit for important social and historical information, and disrupt traditional patterns of transmitting domain expertise.

This article briefly reviews other examples of media serving as catalysts for change and conduits for multiple levels of information. It then describes the domain of classical scholarship, focusing especially on the construction of textual editions and paper-based tools for searching, and the construction of computer-based textual representations and search tools. It applies the package metaphor to outline the way in which the nature and practice of scholarly work is affected by this shift in medium. The article concludes by suggesting implications this research holds for both the academic and practitioner IS communities with respect to the representation of ambiguous and context-dependent data, the construction of tools for data manipulation, and the inte-

gration of both data and “expertise” from a wide variety of sources. These experiences are particularly important in light of the move toward the establishment of digital libraries and the use of gophers and ftp sites to store and disseminate information.

2. KNOWLEDGE WORK AND NEW MEDIA

As computing applications move away from automating routine tasks, interest in the IS community has shifted to supporting complex forms of “knowledge work.” Davis et al. [1991] define knowledge work as consisting of tasks in which the dominant activities include: the generation of useful information, dependence by the individual on accessible knowledge, the use of a mental model of process and output, and the need for significant attentional information processing. The work of classicists fits this description, as does the work of managers, executives, software engineers, and R & D staff. Davis et al.’s interest in a conceptual model parallels systems-development efforts reported in journals such as *Information Systems Research*, *MIS Quarterly*, and *Organizational Computing*, which regularly report on tools to support work that requires “mental” or “intellectual” expertise.¹

The basis for these systems is the *data* that is to be represented and utilized. Documents, artifacts, and “collected wisdom” already exist within groups, organizations, disciplines, or cultures. Capturing, encoding, and using this data in an effective and appropriate way is key to their successful development. The development and deployment of these systems raise questions with respect to user perception of these applications, their inherent usefulness and limitations, and the very notions of “data,” “information,” and “expertise” upon which they are based. Several issues seem particularly important as we look at the shift from paper/books to on-line databanks. First, what data is being captured, and how is it being represented? Second, how do users view the new tool, and how do they understand their effective use? Finally, how do expectation and use mesh with the *actual* capabilities of new tools? Before addressing these issues within the domain of classics, let us consider evidence from other areas that might suggest some answers.

Medium, thought, artifacts, and work processes are deeply intertwined, often in ways we do not fully understand. Henderson [1991, p. 448], for instance, reports on a drafter who claims she cannot think without her traditional drawing board, even when a CAD system would let her create the exact same drawings. An operations manager whom Zuboff [1988, p. 165] interviewed fears that a move from paper ledgers to computers will distance him and his employees from the data they use. Some classicists retain a deep, abiding love for the books they use, while feeling alienated from electronic versions of the very same texts [Ruhleder 1991, p.165]. Another classicist claims that the availability of a wide variety of electronic texts gives graduate

¹Topics covered in these journals include executive information systems, decision support systems, end-user development of decision aids, cognitive fit and information accuracy, data filtering and condensation, modeling, and the use of graphical representations of information.

students a different “perspective” on “what’s out there” [Ruhleder 1991, p. 156].

The computer is not the first medium to provide novel opportunities for the creation, manipulation, and transmission of information. Earlier changes from one medium to the next suggest that a given technology often supports a particular kind of experience or shapes a user’s way of thinking. The combination of skills required to produce and use the artifacts associated with a given medium also defines their integration within a given culture [Clanchy 1979, p. 88]. For instance, the development of stable writing materials, coupled with the spread of literacy, led to the decline of oral culture [Ong 1982], which in turn spurred the development of cursive script for quicker transcription [Clanchy 1979]. Phonetic alphabets (as opposed to hieroglyphs or ideograms) create a sharp division between auditory and visual experience, and support lineal sequences as forms of psychic and social organization [McLuhan 1964]. The advent of the printing press led to reference works with novel layouts, forcing readers to reorganize their thinking and apply new cognitive skills [Eisenstein 1979]. This era, for instance, spawned a series of works based on an alphabetical, rather than thematic, order [Eisenstein 1983]. Nowadays, computerization of a dataset can often be a catalyst for rethinking former classifications [Collins 1990b].

In the future, technical advances in high-resolution graphics and multimedia workstations may make it possible to support a wider variety of cognitive organizations of the same data, and to support a wider variety of forms of data. Some forms and organizations may be particularly suitable for certain purposes. For instance, modern science is deeply indebted to the ability to widely disseminate ideas via two-dimensional, reproducible, transportable inscriptions [Irvins 1953; Latour 1986]. Hypertext may be found to be more compatible with the relational way of thinking characteristic of a technical culture, and hypertext systems may support particular forms of intellectual inquiry especially well [Crane 1988]. Computing developments in the social sciences may eventually help integrate quantitative and qualitative approaches [Ragin and Becker 1989]. In some cases, however, the convenience of a particular technical option may result in an inappropriate representational choice [Clark 1991]. And even if a faithful transference of information from one medium to another were possible, any representation will highlight (or even distort) certain features, while completely omitting others [Monmonier 1991, p. 25]. Moreover, *technical* solutions do not address our limited ability to capture, represent, and/or codify certain forms of data.

Both Collins [1987] and Perrolle [1986] would argue that technical possibilities are moot as long as developers remain mired in an understanding of data and information that is based on “arithmetic” or “mechanical” paradigms of knowledge. Instead, they and others argue that data and information are socially constructed, and can only be understood within a given social context. Expertise within a given domain is understood and passed on through what Collins [1987; 1990a] terms an *enculturational model*, in which knowledge is shared within the social context of a laboratory or other apprenticeship

setting.² In one office, clerks experienced a “loss of concrete stimuli to trigger memory” of dealings with a client, portfolio, or problem when shifting from a paper to an electronic medium. One former stimuli had been the margin note, which helped create a shared history between the document and the individual(s) working with it [Zuboff 1988, p. 132]. Adapting the notion of media richness from communications theory, Daft and Lengel [1984] show how artifacts that are usually considered lean in information content (such as budgets, schedules, or database updates) are made rich by virtue of their situatedness within the organizational context. Their linkages to organizational history, informal knowledge, and personal alliances enable them to be used in settings characterized by uncertainty and equivocality. Finally, not all forms of information are equally reproducible; important information about an artifact may lie in the *process* of its evolution. In tool development, for instance, the *shared act of creation* is as important as the product that results [Bly 1988; 1990].³

Classical scholars epitomize the knowledge worker as defined by Davis et al. [1991]. By understanding the impact of computerization on their domain of work, we will come to understand better this impact on similar domains. One particular tool—an on-line databank of Greek textual sources—will serve here to highlight a variety of phenomena that parallel some of the experiences noted above. Given past experiences with new media, we may expect to find both information losses and gains as this single, computer-based source is constructed from multiple paper-based sources. We may expect to see the relationship between the tool and the user change as a new representation (via a new medium) alters the user’s perception of the tool, its validity, and its utility. And we may anticipate a rupture in the traditional transmission of domain expertise from one generation of classicists to the next as the necessary skills for “doing scholarship” and “using the tools of the trade” continue to change.

This article is derived from a broader empirical investigation of the impact of information technologies on classical scholarship [Ruhleder 1991]. It draws on approximately 60 unstructured interviews conducted over the course of 5 months with faculty in Classics Studies departments, tool developers, and editors of traditional journals and electronic mailing lists. Interviews were conducted in person whenever possible; data were also gathered via telephone and electronic mail. These interviews were supplemented by limited participant observation and the use of secondary sources, including journals, newsletters, and conference proceedings. The original investigation consid-

²See Collins [1990a], Brown and Duguid [1991], Lave and Wenger [1991], and others for a more extensive discussion of this topic.

³Similarly, a work of art has a unique “presence in space and time,” and the “information” it conveys depends as much on the viewer and the setting at the moment of viewing as on the artifact itself [Benjamin 1968, p. 220–221], just as the “information” conveyed by a musical performance is a complex interaction between the musical “artifact,” the performers, and the audience [Cage 1961].

ered a variety of information technologies, including word processors and electronic mail.

This article focuses specifically on the integration of one computer-based artifact into the discipline, and on the changes that this integration has occasioned. It employs the metaphor of an artifact as a complex “package.” First articulated in Kling and Scacchi [1979], it defines *computing packages* as consisting not only of hardware and software, but also of skill sets, data, beliefs about computing, and organizational support infrastructures. Similarly, both computer-based and traditional artifacts classicists use to carry out work can be described as complex “packages” embodying expectations about the work process, the expertise needed to carry out work, the values of those who use them, and the existence of an infrastructure necessary to create and maintain them. We return to this metaphor after a description of the domain of classical scholarship.

3. CLASSICAL SCHOLARSHIP AND TEXTUAL (RE)PRESENTATIONS

*We preserve the past because we want to
know who we are, where we come from.
Culture, continuity—this is like knowing
your family's past. It's a human need.*

—Classicist and TLG Staff Member

Classical scholarship traces its roots back to the 8th century B.C., and was already an established activity by the 3rd century B.C. [Reynolds and Wilson 1974]. The goal of modern scholarship is to bring to life “Graeco-Roman civilization in its essence and in every facet of its existence” [von Wilamowitz-Moellendorff 1982]. Scholars rely on various forms of textual evidence, including papyrus fragments, medieval manuscripts of ancient texts, and modern editions in conventional book form. They use these materials to debate questions of manuscript authorship, evaluate the relationship between different groups or classes of people in ancient Greek society, or investigate different meanings of a particular word. For instance, one classicist published a book that explored the Greek tragedians’ use of allusion to evoke in the listener the emotional context of other works. Scholars test their hypotheses about a text by locating potential pieces of evidence within other texts and then applying analytical techniques to weigh and interpret that evidence.

Two particularly important kinds of artifacts are textual editions and tools for searching (particularly concordances and indexes). Textual evidence is an important source of discovery about Greek and Roman civilization and an equally important source of support for evidence gathered by other means, such as archaeological digs or works of art. But to talk about “the text” is deceptive; any scholar working with ancient Greek and Roman writings knows there is no such thing as a definitive text. Instead, one works with materials of a fragmentary and questionable nature. In many cases, only copies of other copies exist, some dating back only as far as the Middle Ages

and subject to scribal error and emendation.⁴ The reconstruction of the original, or Urtext, is an important primary activity within classical scholarship.

By the time a particular text appears in book form, the editor of the edition has made a series of decisions about variant readings and about the inclusion or exclusion of certain questionable sections of text. Sometimes, for example, the original author might have produced multiple versions of a text over the course of his or her life and the editor must decide at what point distinct versions of the text can be identified. These kinds of problems and considerations are not limited to classicists, of course, but affect any who are working with similarly fragmentary or questionable textual data. While no text is free of such problems and issues, one of the differences between various presentations of the text lies in how *implicitly* or *explicitly* these decisions are represented in the textual “package,” and how the various critical materials are physically coupled to the text itself. Different constructions of the “package” may have ramifications for the work of classicists in terms of problem choice, work strategies, and their conceptualization of the text and corpus of texts. The same is true of search tools, both traditional and computer-based. The ways in which these “packages” are constructed are outlined in more detail below.

3.1 In the Classical Tradition: Textual Editions and Tools for Searching

“[You find a word or phrase that looks like an allusion and ask,] where else might that word or words be? And there comes the rub. That would then entail [for each] author, say Homer, taking the index or concordance to Homer, and checking how many occurrences of that word or phrase there are. . . . But then you need to check other places, so then you check, is it a rare word? You start out with a dictionary. Sometimes that will indicate it to you, but then, basically, you have to go to individual indexes and concordances for each author, and just see where all it’s used, how it’s used, if it’s used very often. It’s a laborious process, you can only play out a certain number of your hunches, go with the best ones.”—Classicist working on allusion in Euripides’ *Alcestes*

Various formats exist for capturing the fragmentary and disputable nature of classical texts, and some methods for marking missing, questionable, or spurious sections of text had already been standardized by the third century B.C. [Reynolds and Wilson 1974; Youtie 1974]. These markings, as well as other forms of commentary or critical notes, form a part of the printed books most scholars rely on when working with a given text. The introduction to a new edition will typically explain the methods or criteria used by the editor in the editorial process, and will place the new edition within the established literary tradition. One special form of critical note is the *apparatus criticus*. This set of textual notes lists variant readings in all or at least the major alternate manuscripts or editions, thereby tracing the history and logic of

⁴Corruptions can be attributed to careless copying, handwriting variations, or merely tiredness on the part of the scribe. Sometimes, however, sections were deliberately omitted or modified for religious or political reasons.

editorial decisions—vital in a field where the family tree or “textual tradition” of an author’s work forms a major component of information about that work.⁵

To use these texts effectively for literary criticism, scholars need to be able to locate words, phrases, and grammatical or metrical forms to support or dispute particular points. There are several basic tools that have traditionally allowed a scholar to search for words and phrases within a given text. These include indexes, concordances, dictionaries, and of course, the human eye. Concordances are particularly useful, and differ from indexes in that they not only indicate the location of a word within a text, but tend to include some context for the word, perhaps even a definition or translation. Because the selection of words included in a concordance is up to the editor, their comprehensiveness and accuracy vary. As with textual editions, tools for searching are generally created specifically for one work or one author’s canon (i.e., Rengstorf’s [1973] concordance to Flavius Josephus). Dictionaries or lexicons, on the other hand, span the vocabulary of numerous texts and authors, and will both define a word and give information about its history and usage.

3.2 The Thesaurus Linguae Graecae: Repository and Search Tool

“Now once the Ibycus [TLG] came along, ... [I could] put up an all-Greek search on the screen, type in that word, tell the machine to search through, and then just turn my back and go about my work, and 40 minutes later I’d look, the machine has a match for me. It will have that match from the Odyssey, and if there are any others, they’ll be there. And if there are no others, it will tell me there are no others. And, with only the amount of time it took me to type in the word and set the machine going, I’ll have the answer.”—Classicist working on allusion in Euripides’ *Alcestes*

The earliest electronic versions of Homeric texts were created in the 1960s using mainframe computers, and scholars continued to encode individual texts for use by themselves or a limited number of colleagues. None of these early on-line, individually constructed text versions have had the impact on the discipline that can be attributed to the *Thesaurus Linguae Graecae* (TLG). This Greek textual data bank was begun in the early 1970s at the University of California, Irvine, by Theodore Brunner (Department of Classics), with the assistance of David Packard, a classicist and son of one of the founders of the Hewlett-Packard Corporation (an engineering firm and com-

⁵The textual edition can also be viewed as a *genre of organizational communication*, as articulated by Yates and Orlikowski [1992], in that it is a typified communicative action invoked in response to a recurrent situation and based on established practices, social relations, and communication media, and is characterized by specific substance and form. Textual data banks may be emerging as a new genre of communication not only within classics, but within the humanities in general, though at this point they fail to meet the criteria as defined by Yates and Orlikowski [1992]. Similarly, electronic journals, bulletin boards, and mailing lists may characterize the emergence of new genres of communication that transform, supplement, or supersede traditional academic journals and newsletters. The author plans to explore this further in a subsequent paper.

puter manufacturer). It currently contains electronic versions “of all of ancient Greek literature surviving from the classical and postclassical periods” [Brunner 1991], and has begun to incorporate medieval Greek texts.

The TLG is the only widely used, comprehensive textual databank for ancient Greek texts. It is well respected in the classics community and has developed a strong infrastructure for acquiring, encoding, and disseminating texts. Conceived as a data bank for Greek literary texts, other materials (such as Latin texts and other Greek materials) are becoming available through the TLG infrastructure. The TLG texts are accessible on CD-ROM, or by using the Ibycus, a dedicated minicomputer that incorporates the TLG texts, a set of search programs, and a word processor for use with Greek texts. Several search packages, developed by other classicists independently of the TLG project, are available for use with the CD-ROM and IBM-based or Macintosh microcomputers. Search results can then be transferred into several popular word processing programs, such as WordPerfect or Microsoft Word. More information about the TLG can be found in Brunner [1987; 1988; 1991] and Ruhleder [1991].

The TLG serves two important functions. First, it allows scholars to access Greek texts selectively, easily, and quickly through a series of on-line search programs. Second, the project itself serves as a central repository of ancient Greek textual materials. The collection of electronic texts is catalogued in the *Thesaurus Linguae Graecae Canon of Greek Authors and Works* [Beckowitz and Squitier 1986], produced by the project. This work attaches a genre to each text and provides information for looking up references in the text found with the TLG. Scholars can search through all texts contained in the TLG, or they can narrow a search by creating a list of specific texts of interest. This can be done by selecting texts individually or by specifying the texts of all lyric authors or all Greek tragedians using the genre specifications of the *Canon*. Once the range of texts is set, the word or phrase of interest is entered, and the TLG searches for a match. If one is found, it is displayed in boldface with three lines of context: the line on which it occurs, the line just before, and the following line. Again, an “index” into the original text is also provided so that the user can locate the phrase within the text itself. Search results can be printed and/or saved to a file if desired. The texts themselves are represented using a special encoding scheme for representing Greek characters and diacritical marks, but none of the critical notes from the textual editions are included.

4. UNPACKING THE PACKAGE

With this overview of both the traditional and computer-based texts and search tools available to classicists/users, let us examine these “packages” both with respect to their ramifications for classicists, and with respect to the issues of data representation and manipulation. To reiterate, the artifacts classicists use can be defined as packages that have the following characteristics:

—they embody some particular form, organization, and content;

- embedded within them are beliefs about the artifact and expectations about the work process;
- they require special skills sets or expertise to understand and/or use them effectively;
- and they assume the existence of infrastructures to create, support, and maintain them.

Many of the changes that are or will be effected by the shift to computerized textual data banks can be expressed as changes to the package, or as the interaction of changes to one or more aspects of the package. In all fairness, the TLG and other textual data banks have helped alleviate many problems faced by scholars in classics and in other academic disciplines. The gains in breadth and search scope alone are staggering when one compares the TLG to traditional tools. The greater access to texts can balance out the relationship between poorer and richer departments, between scholars at the beginning of their careers and those at the end. And yet, many claims about the TLG's utility embody some level of false assumptions or misunderstandings about the system itself, and its effective use will require major changes in current educational and tool-building infrastructures. Examining these aspects will help us understand the broader ramifications of changing the traditional "package."

Definition and Standardization of the Canon. The works of classical authors have traditionally been scattered throughout libraries and archives. In fact, constructing a bibliography for a particular author or set of authors was once considered a scholarly contribution. As a repository, the TLG has achieved an important milestone in classical scholarship: the establishment of and standardization of a canon of Greek texts. According to the TLG's director,

"In 1972, '73, nobody knew who had written what, when... We had to find the texts to each one of those authors. In the case of Homer or Plato it's easy to find the best edition in the bookstore. But there were many, many authors which were out of print, of whom only one edition was cited somewhere. We went as far as Czechoslovakia and hand-carried texts back. A few weeks ago [a former graduate student now at the University of Crete] ran into an edition that was published around the late '50's... and it's now on its way express mail to UCI."—Brunner, TLG Director, 1990 interview.

The collection of electronic texts and the attendant *Thesaurus Linguae Graecae Canon of Greek Authors and Works* is thus of fundamental importance to the discipline. The *Canon* and TLG serve as physical embodiments of the very idea of a single corpus of ancient Greek authors. The selection of texts is carried out by a committee of the American Philological Association. Unfortunately, the choice of a particular text is often disputable. One classicist explains,

"The version of Euripides the TLG put in is Murray's text, which is 80 years old, now two-thirds replaced by a newer edition by a Cambridge scholar. That's what people now use for the authoritative text. You might do a search on the Ibycus and find something interesting, and then open up your newer edition and see that it was just Murray's choice, and Murray was wrong."

Genre tags, attributions of texts to authors, and other information are also open to dispute. Eventually, the TLG hopes to include all versions of extant texts⁶ and an *apparatus criticus*, but any classification will remain open to criticism in a field where multiple interpretations and codifications of information have been the norm. However, the TLG is becoming such a dominant resource that it is not clear how well these alternative viewpoints will be preserved over time, a topic taken up again below.

Democratization of Access and the Role of Infrastructure. Until recently, scholars relied on the library of their home institutions, interlibrary loan arrangements, or travel grants to gain access to source materials for their work. Library quality constitutes one factor in ranking a department, and younger programs trying to build up a library find that many needed texts are out of print. With the TLG, the user has access to more source texts than libraries can provide. This not only opens up the possibility of addressing questions requiring broader textual access, but allows a savvy user to exploit new opportunities for extending previous research. (A graduate student recently overturned established wisdom about the meaning of the word “disciple” at the time of Christ with the help of the TLG [Wilkes 1988].)

However, what some have labeled a democratizing or equalizing process also raises questions with respect to resource allocation and infrastructure building. In order to access the texts of the TLG, individuals and/or departments have to make significant investments in computing to secure the needed hardware, software, and licenses [Ruhleder 1995]. An administrative assistant in a medium-sized classics department explains the trade-offs:

“We’ve had to reallocate, but we haven’t gotten any more [funds] for reallocation.... You have to whittle down some costs in order to pay for others, and that’s what this department has done. [Costs such as] annual licensing fees, those have been absorbed by other funds in the department.”

In this case, monies were drawn from an endowment fund earmarked for “general use,” from which were also drawn funds for the classics library and for students to attend conventions. In older buildings, merely finding the money to lay down electrical lines can be difficult as it is not typically considered a legitimate academic expense; at the time of this study, at least one major classics department at a prestigious university did not have any email connections for this very reason.

The traditional “packages” of textual editions and search tools bear a certain legitimacy as tools of the academic trade. The infrastructures both for their creation and acquisition are firmly in place and have been for some time. An infrastructure supporting the creation and dissemination of computer-based tools is developing slowly, and is often still quite project-specific.

⁶A few years ago, the TLG began computerizing medieval Greek texts. Some classicists criticized this decision, saying they would have preferred the TLG to work on including additional versions instead. Some attributed the decision to the fact that it is easier to get funding to encode new works than to encode additional versions of texts already on-line. The TLG staff, however, disputes this as the rationale for the decision.

Twenty years ago, the TLG had to create its own support mechanism, including the privately funded Packard Humanities Institute. Other projects, such as the Perseus Project and the Text Encoding Initiative (TEI), now receive outside funding and the legitimacy that comes with it.⁷ Newsletters put individuals in touch with electronic sources of information.⁸ Yet until computer-based tools gain legitimacy not only within but outside of the discipline (i.e., on university budget committees), it will be difficult for individuals to develop *local* infrastructures for their acquisition, maintenance, and continued use.

A related access issue is that of training or expertise. From an infrastructural perspective, new tools necessitate new mechanisms for training people to use them. The use of traditional tools is already a part of undergraduate and graduate curricula classics, passed on through an apprenticeship model of learning. It is less clear how training on computer-based tools can be effectively integrated into extant curricula, especially when those possessing the domain knowledge (classics faculty) may have extremely limited technical knowledge. Some graduate programs now include a one-time seminar “using the Ibycus,” while at least one university has a full-fledged certificate program in humanities computing [REACH 1991]. This issue is addressed below from another perspective.

Breadth of Access and Changes to Work Practices and Assumptions. The TLG allows classicists to use a much wider set of materials in their work. According to one classicist,

“[To see] the range of texts that are out there, and sources, that’s almost an education in and of itself. . . . I think it really does give people a sense of a broader horizon. . . . They see all these medical writers coming up [in a search]; they wouldn’t have thought about medical writers, late grammatical writers.”

The question, of course, is what exactly one should *do* with all these extra materials. They allowed the graduate student referred to above to do a more complete analysis of the use of the word “disciple.” It is not clear, however, if this broader access would have helped him come up with the original project, conceived 40 years earlier by a senior scholar, then at the apex of his career. While one classicist believes that using the TLG “will force people to ask

⁷The Perseus Project, a hypertext/multimedia tool for teaching and research across many subdisciplines of classics, receives major funding from the Annenberg Foundation; contact Dr. Gregory Crane, Director, Perseus Project, at the Department of Classics, 319 Boylston Hall, Harvard University, Cambridge, MA 02138. The TEI is developing a generic encoding scheme for all languages and is funded by the National Endowment for the Humanities, the Andrew W. Mellon Foundation, and the European Economic Commission; contact Nancy Ide, Steering Committee Chair, Computer Science Department, Box 520, Vassar College, Poughkeepsie, NY 12601.

⁸REACH, Research and Educational Applications of Computers in the Humanities, the Newsletter of the Humanities Computing Facility of the University of California at Santa Barbara, Santa Barbara, CA 93106-3170, is one such source; contact Eric Dahlin, hcf1dahlin@csbuxa.ucsb.edu.

more legitimate questions,” another observes:

“[I see people] printing up a big search and printing it out and feeling like they’ve done something when they really haven’t done anything.... That doesn’t mean they thought about anything or read anything or looked at anything, and you can see that as kind of a crutch and a substitute.”

At the same time, breadth of access is changing the definition of good or complete work as it redefines people’s sense of what amount of work an individual can do in a given span of time. This includes an idea of a reasonable scope for a dissertation, or the number of authors or works one needs to consult in order to make a definitive statement. Certain types of word studies are no longer acceptable, and journal editors are sending authors back to the computer as using the TLG becomes the mark of “doing complete work.”

The use of the TLG may also affect an important assumption about the work process: the notion that statements about a text are based on an intimate knowledge of the text. Greater access to textual material in electronic form increases the range of texts one can use in one’s work without having read them thoroughly. When asked what difference the lack of an Ibycus would make to his dissertation work, one graduate student replied:

“It would make [the dissertation] far more labor intensive. I could do it, but I would feel far more diffident. I would have to read... through the orators, which is a pretty large corpus, and the philosophers, which is huge.”

He *is*, however, using textual materials from the orators and the philosophers gathered via the TLG to construct the arguments for his dissertation.

Examples such as these lead established classicists to see the TLG as interfering with the traditional relationship between text and scholar, of eliminating the intensity of personal discovery, and of removing the serendipity factor (also a problem with on-line card catalogues, and filtering systems for electronic mail or news). A papyrologist states, “It gives you a real sense of what X was like to have read it yourself,” while a latinist adds, “The basis is reading texts. When you read, you learn things that you aren’t even searching for. The human brain may be searching for X, but will be confronted with other things.” Finally, one historian who has limited access to the TLG concludes, “I find it quite salutary not to have a computer [because] I sit down and read.” To some extent, this is a generational factor, since most of the people who are concerned about the role of computing are older and established. It is not clear what role the use of the TLG and similar tools will play when the “younger generation” reaches their level, or if these concerns about depth will be handled by the traditional review process.

Critical Notes as a Shared Act of Creation. As mentioned above, the TLG also leaves out a traditional component of the textual edition, the *apparatus criticus*, described above. In doing so, it eliminates a complex intellectual tool that carries important links and historical markers. A classicist explains:

“Creating an *apparatus* is a very labor-intensive and intelligence-intensive process if you want to do a good job at it. Because of the way the apparatus is written

in relation to the text that's above it, it takes some intelligence to decide what to put in your version. If editor A puts a particular word into the text, then what he puts in the apparatus may be the very word that is already in [the text of] editor B. And for some authors there's so much controversy over the text that whole phrases are in question. You have to make a decision: how big a phrase will you put into your *apparatus*? Are you just reporting a single word? In some cases, that single word will be incorporated into a larger word in a conjecture. And so that, so a person who's putting things together like that has to know something about what they're doing."

Physically coupled to the text itself in the traditional edition, the *apparatus criticus* serves as a vivid reminder to scholars that the text before them is one person's "best guess" (at one particular point in time) at an *Urtext* which can never fully be recovered or reconstructed. The lack of critical materials deprives the scholar of an important set of links, since an edition is essentially a representation of information about a text, including its history and its reliability as a witness to the archetype. The interpretations of a text are their local instantiations, and these interpretations over time serve as their "re-representation path" [Star 1990].

The apparatus and other tools, such as concordances or dictionaries, ground the reader/user in the social context within which the text or tool was constructed, providing the individual with access to a collective expertise developed literally over centuries. When one classical philosopher describes his work as a "cumulative, collaborative effort," he means an effort to which contributions have been made "by scholars over the generations." The *apparatus* and other elements of traditional packages capture this collective expertise. While these links are still available, they must now be recovered by the individual from another repository altogether. We can visualize the shift from one "package" to the other as presented in Figure 1 and Figure 2.

The new form of textual "package" as embodied in the TLG changes the picture the scholar has of text and corpus. The view is flattened: broader, in that one can "see" more texts from where one stands, but also shallower, as critical information is decoupled from the tool at hand.

"Objectivity," "Accuracy," and the Shifting Locus of Expertise. Two oft-cited benefits of the TLG's computerized search programs are its "objectivity" and its "accuracy" with respect to searching. The accuracy of the TLG in locating specific search patterns, coupled with the breadth of texts available through this data bank, allows scholars to attack a problem with a thoroughness they might not have achieved before. According to one classicist:

"You could never be as confident that you were covering things as well [as you could with a computer] because some of the concordances are way out of date, and even the most up-to-date printed Greek dictionary is designed on certain principles that are not designed for exhaustiveness."

Furthermore, the tools and techniques for searching the TLG texts are standardized, in contrast to the wide-ranging quality of traditional tools.

However, these attributions of "accuracy" and "objectivity" to the TLG highlight an interesting interaction between assumptions about the medium (the computer) and the lack of computer-specific expertise required to under-

Fig. 1. Traditional textual edition: Single text coupled with critical materials.

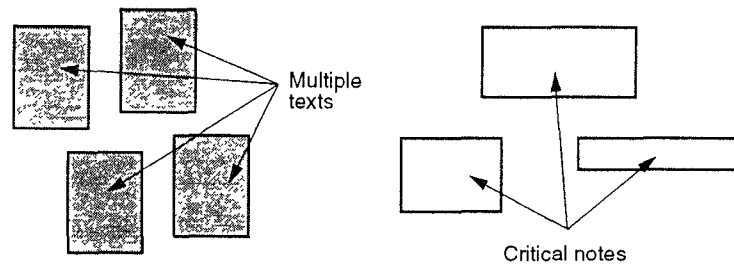
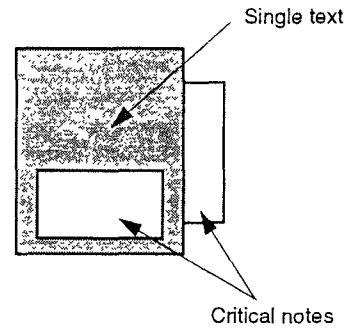


Fig. 2. Computer data bank: Many texts decoupled from (paper-based) critical materials.

stand the limitations of computer-based tools. According to one individual, “[The TLG] is an objective program, it only looks for what you tell it”; indeed, the TLG will unfailingly identify each and every instance of a particular search pattern. Yet even as they complain about the lack of an *apparatus* or multiple versions of a text, few classicists seem to recognize the extent to which domain expertise is no longer embedded within the tool, and what this might mean in terms of the *correctness* of results developed by using the TLG.

For example, when searching for a particular word or phrase, the scholar must be aware that spelling might vary over time, by place, from scribe to scribe, or even within a text created by a single author [Reynolds and Wilson 1974]. Without knowing all the possible variations, one cannot effectively use the TLG to “accurately” and “objectively” identify all locations of a word or phrase. Someone working with the corpus of fifth century decrees faces this problem: “It’s a little bit tricky to use [the TLG] because the orthography is in flux at this point.” In contrast, the editor of a traditional concordance, presumably an expert with respect to a particular text and era, would have indicated orthographical fluctuations in some manner. Even if this were not the case, similarly spelled words will still be located near one another, increasing the likelihood that the scholar will discover a close variant. In the case of the textual edition, the *apparatus* might contain notes about the orthography, or about variants in other manuscripts. The standardized na-

ture of the TLG search programs and their application, while providing more extensive search coverage, puts the burden of recognizing the need for, obtaining, and appropriately using this critical knowledge on the user/scholar. This is but a small example of how what was once the responsibility of an editor is now the responsibility of each individual database user.

Access mechanisms and forms of expertise also interact since certain kinds of information can only be retrieved through the use of computing technologies. In order to make effective use of their *domain* expertise, classicists must develop the *technical* expertise required to correctly use genre tags, to construct lists of authors for searching, and to develop complex search patterns. To do this, users need a set of skills specific to use of the TLG and the *Canon*. And, as fewer individuals or departments invest in an Ibycus stand-alone, users are required to develop enough general computing skills to set up and use a CD-ROM, the search package developed for use with their IBM or Macintosh, and then save and retrieve the results of a search and incorporate them into their own writings.

Even if they possess both the domain and technical knowledge to use the TLG effectively, few classicists have the knowledge of computing that would enable them to understand the limitations—and the subjectivity—of some aspects of the TLG's creation. The texts are entered and verified according to a set of programs and algorithms developed by the TLG staff. When these algorithms cannot handle a situation, TLG staff members must resolve problems by making decisions about a particular reading. As mentioned above, works are also “tagged” or labeled according to various characteristics, such as genre, which are open to dispute. Use of the *Canon*'s tags to construct search lists may mean the user ends up with a different set of authors in the search set than anticipated. Furthermore, these searches rely on a set of programs developed by the TLG's staff, which may contain errors or problems that will only be identified by continued use. Finally, as individuals not affiliated with the TLG develop search programs for use with other personal computers, the quality controls provided by the infrastructure and expertise of the TLG project is eliminated.

In contrast to a textual edition with explanatory notes in the introduction, these complex decisions about the construction of the TLG texts and search algorithms are hidden from view because of the manner of its presentation. Even if they were presented up front, it is not clear that this could be done in a way which would be comprehensible and useful to the scholar/user community at large. The construction of a concordance or dictionary is easy for a classicist to understand; this is not true of a set of computer-based verification and correction programs for encoded textual data, or the code of a search procedure. In classics and in similar settings, an increasing amount of technical expertise is necessary not only to use tools, but to understand their creation and to potentially challenge decisions about their construction.

5. IMPLICATIONS FOR CLASSICS

For classicists, the shift in medium from traditional to computer-based tools has mixed implications. The advantages of the resulting “package” parallel

experiences with earlier forms of new media or within other domains of computerization. The construction of the TLG has helped further stabilize, preserve, and disseminate a far wider range of texts than any single library has been able to collect to date, just as the printing press did five centuries earlier. Both the TLG texts and search programs serve as “reproducible, transportable inscriptions,” ready for uniform local instantiation in multiple work settings, opening up to classicists some of the benefits that the sciences reaped by means of such inscriptions. Yet the TLG has also served to decouple the text from important critical materials that reflect the history of its transmission and the process of its construction. Its technical base has changed the expertise needed to understand, use, and construct tools. This, in turn, has interrupted the traditional “apprenticeship” model of the transmission of domain expertise. We can capture some of the changes and their implications by reviewing the characteristics of “tool packages” used by classicists, and the shift from the traditional to the computer-based.

Packages embody some particular form, organization, and content, all of which have changed with the introduction of the TLG and similar tools. The change in *form* is the most obvious; the organization and content have shifted from a combination of context-based source and secondary materials to a linear stream of encoded textual information, while at the same time standardizing the corpus. Embedded within these packages are beliefs about artifact and work process. While there has been no change in beliefs about how a scholar *ought* to work, the tools no longer support a process that relies heavily on contextual linkages in a “collaborative process” that literally spans centuries. Packages require special skills or expertise for effective use; traditional “tool packages” have embedded within them significant domain expertise. This has now been stripped to a large extent from the computer-based version of the artifact, which now also requires an additional layer of technical expertise for effective use. Finally, centuries of use and practice created infrastructures for the development, evaluation, and dissemination of textual editions and concordances; these were integrally linked to curricula in classical studies. No such infrastructures exist as yet for the development and evaluation of computer-based tools, and for their incorporation into traditional curricula.

6. CONCLUSIONS

The issues raised above are also important to other knowledge work venues. We are still making a transition from an era in which the focus of software development was on transaction-processing systems used by a limited number of people, to an era in which the focus is on systems to support individuals and groups working within complex organizational settings on tasks we consider “knowledge work.” What lessons can we draw from this study to help us pursue and evaluate developments in this arena? And what aspects of the package metaphor can be particularly useful in helping us understand their ramifications?

First, let us consider the notion of “system success” with respect to the kind of data or information provided to the knowledge worker. The TLG is a success in that it is widely used and provides data not previously available. Yet from the perspective of the package metaphor, it is not a success in its ability to provide its users with the kind of information they need in their work because it fails to embody important beliefs about the original artifact (the textual edition) and expectations about the work process (textual criticism). It has *decontextualized* the data critical to the user’s needs by stripping the “raw data” (the text) of its historical or organizational context. This is precisely the same shortcoming that Henderson’s drafters identified in the move to a CAD system, and which Zuboff’s managers complained about when their ledgers were computerized: “Some ledgers have ten years of history written in them” [Zuboff 1988, p. 165]. Information and its use in decision making is highly contextual and resides not within individuals or single artifacts, but in *communities-of-practice* [Brunner 1987]. “Successful” development, using the richer criteria of the package metaphor, requires the capture of ambiguous, often inarticulable, forms of information, embedded in communities of practice which extend across traditional organizational boundaries.

Second, one key component of a package is the special skill or expertise needed to use an artifact appropriately. When complex artifacts are used, one way of developing that skill or expertise is for users to play an integral role in the development process. While tool building is still an integral part of work in some professions [Collins 1990a; Traweek 1988], increasing numbers of systems that support professional activities are being constructed by software developers or “knowledge engineers” without much (if any) domain expertise. It is often no longer possible for users to understand the construction—and hence the limitations—of the tools they employ. For instance, most classicists have only a rudimentary understanding of computing, yet are willing to attribute an unwarranted accuracy and “objectivity” to computer-based results. In a more dramatic example of overconfidence in computing, patients burned by the Therak-25 radiation machine were told they were mistaken because “the computer” *must* have given the correct amount of radiation [Leveson and Turner 1992]. Leonard-Barton and Sviokla [1988, p. 97] caution that users expect higher, more consistent performance from expert systems, but that “such lofty expectations court disaster,” particularly when the “correctness” of an answer is a matter of judgment and opinion. And Will [1991] documents the fact that even experts in a given domain will not necessarily question *incorrect* results when using an “expert” system. While it is not always necessary to fully understand the construction of the artifacts we use (the automobile is an oft-cited example), we do expect experts to have a deep enough understanding of the tools they employ to use them appropriately and effectively within their own domains.

Systems such as the TLG may challenge or stretch complex infrastructures for the transmission of knowledge. Entering a profession still embodies aspects of traditional apprenticeship arrangements [Collins 1990a]; learning and problem solving occurs in the context of actual practice [Brunner 1987]

(though a new tool can sometimes enhance cohesion by fostering a common language [Sviokla 1990, p. 137]). The shift to systems that are not understood or effectively used by “older” members of a professional community disrupts the traditional transmission of domain expertise, which goes beyond imparting “facts” to teaching newcomers how to locate, evaluate, and integrate many forms of data using a wide array of tools, techniques, and strategies. Furthermore, systems such as the TLG throw open the very question of who is able to carry out a particular kind of work. Crane [1988] suggests computer-based tools in the Humanities will allow individuals to address questions outside their own domains of expertise. Yet these individuals would be even less likely to understand results generated by the tool, a development against which Sviokla [1990] warns with respect to the organizational deployment of expert systems.

These problems reflect the conflicts between “arithmetic” or “mechanistic,” and socially constructed or “enculturational” paradigms of information and knowledge discussed above. Some of them may be addressable via a set of technical solutions, yet applying the package metaphor indicates that solutions focused only on technical aspects are of little value here. The Perseus Project, for example, combines hypertext and multimedia to supplement and enrich textual materials from the TLG with commentary, maps, and photographs in an attempt to better support established information needs and work practices. Yet conversations with the developers indicate that infrastructural issues such as contributions from the discipline, legitimation within the field, dissemination, and use, etc. have not yet been satisfactorily addressed. At the same time, courses and certificates in humanities computing that incorporate systems such as the TLG are raising the technical skill levels within the community, yet they do not necessarily convey the skill needed to understand the construction of systems such as the TLG, or the nuances of their use.

Systems developers in other venues are grappling with similar issues. Developers of the Worm Community System (WCS) [Schatz 1991–92], a workstation designed to support geneticists working with the nematode *c.elegans*, want not only to capture and codify data to be shared among members of the community, but also to support and enhance open and free communication between members. Problems related to skill levels and the development of technical infrastructures continue to limit its use within labs, while increased competition in a growing field works against its loftier goals. The Virtual Notebook System [Gorry et al. 1988] is pursuing more modest goals while building an on-line, multimedia lab notebook workstation; designers of this system recognize that some forms of information may never be captured, and others may only be incorporated through pointers to information and expertise that lie outside the system. These efforts will provide us with some of the skills and understanding needed to create artifacts rich in contextual linkages, represent ambiguous data, and support complex forms of knowledge work. However, we cannot forget that they are taking place in academic atmospheres that support long-term, iterative development, a luxury not shared by commercial development efforts.

It is also not clear whether any of these systems will ever become fully integrated within their own respective disciplines. In fact, despite advances in technologies to support hypertext and multimedia (making it easier to represent multiple viewpoints and diverse information linkages), the actual trend is in the opposite direction. The growth of the Internet, and the emergence of electronic interfaces (such as gopher, archie, WAIS, etc.) to help users locate, browse, and retrieve electronic documents, are changing the overall “information retrieval picture” [LaQuey and Ryer 1993]. The organization of information is similar to that of the TLG, since it tends to be textual in nature, and linear in its representation. Tools such as the TLG or the WCS have at least one advantage: a relatively small, fairly homogenous user community with a shared set of assumptions, motivations, and needs. Even so, WCS is losing ground as geneticists prefer “plucking information off the net” instead of investing in an expensive, cumbersome workstation. Emergent digital libraries and information services must serve a user community unlikely to make heavy investments either in special computing equipment or in the time required to learn interfaces for information services that may be used only sporadically. Even awareness of the range of information available is a major educational challenge.⁹

The growth of networks and on-line information services also highlights questions of information value and integrity. Materials systems such as those that the TLG and WCS include are constructed and approved by complex infrastructures that developed prior to, and independently of, the systems themselves. By rigidly controlling contributions, developers have been able to use established infrastructures to control content and quality. Information dispersed via an open network such as the Internet has no such checks built in; all information is displayed in an identical manner. Without extensive domain expertise and *social* expertise (to what community does this person belong, and what value do I attribute to a contribution from that sector?), it is difficult, if not impossible, to assign value to electronic information. Discussions about allowing users to annotate or informally publish materials within Perseus or WCS raise similar issues. More structured and tightly controlled versions of the digital library may solve some of these problems, but the very nature of the medium seems to fight against this kind of alternative.

Systems such as the TLG provide an interesting intellectual testbed for examining the complex packages that support our everyday work—and the disruptions that occur as we shift the form of these packages from paper-based to electronic media. As our reliance on electronic mechanisms for storing, retrieving, and disseminating information increases, we need to evaluate carefully how this new medium affects the way that we access this information, and the value we place on it. In the case of Classics, the move to a computer-based system that “replaces” traditional textual editions and concordances has affected all aspects of the package, including not only form but

⁹A query to soc.culture.german (USENET), for instance, pointed me to a gopher at the Technical University of Munich listing new German postal codes. Required: several software packages—and recognizing the Internet as an information source.

content, work practices, requirements for expertise, and infrastructural arrangements. These experiences parallel both earlier shifts in media information storage and dissemination, and current systems-development activities in a variety of domains. In the course of altering some aspects of the package, information is lost to knowledge workers as key forms of materials are decoupled or omitted, and work practices and assumptions are reshaped. No matter what the venue, these conclusions have ramifications both for the developers of systems to support knowledge work and for the users who rely on them. Furthermore, they highlight some of the challenges posed by the growth of the Internet and its opportunities for electronic information retrieval from a diversity of sources.

ACKNOWLEDGMENTS

Many people contributed, but I especially wish to thank my colleagues Dennis Adams, Carolyn Green, Rudy Hirschheim, John Leslie King, Wanda Orlikowski, and Susan Leigh Star for comments on various versions. Jeannie Ross served as the "voice" of the IS community, with whom I discussed the ideas on moving from transaction-processing to knowledge-based systems; and the careful critiques and commentaries of four anonymous reviewers helped me to both develop this article and supply ideas for further exploration and analysis. Rob Kling, editor of this special issue, has been extremely helpful with his suggestions and guidance. I appreciate his help and patience. Finally, special thanks to Christine Scott for painstakingly proof-reading the final version.

REFERENCES

- BENJAMIN, W. 1968. The Work of Art in the age of mechanical reproduction. In *Illuminations*, H. Arendt, Ed. Schocken Books, 217–251.
- BERKOWITZ, L. AND SQUITIER, K. A. 1986. *Thesaurus Linguae Graecae Canon of Greek Authors and Works*. Oxford University Press, New York.
- BLY, S. A. 1988. A use of drawing surfaces in different collaborative settings. In *Proceedings of the Conference on Computer-Supported Cooperative Work*. 250–256.
- BLY, S. A. AND MINNEMAN, S. L. 1990. Commune: A shared drawing surface. In *Proceedings of the Conference on Office Information Systems*. ACM, New York, 184–192.
- BROWN, J. S. AND DUGUID, P. 1991. Organizational learning and communities-of-practice: Toward a unified view of working, learning, and innovation. *Org. Sci.* 2, 1, 40–57.
- BRUNNER, T. F. 1991. The *Thesaurus Linguae Graecae*: Classics and the computer. *Libr. Hi Tech* 9, 1 (Sept.) 61–67.
- BRUNNER, T. F. 1988. Overcoming 'verzettlung.' *Humanities* 9, 3.
- BRUNNER, T. F. 1987. Data banks for the humanities: Learning from the *Thesaurus Linguae graecae*. In *Schol. Commun.* (Winter).
- CAGE, J. 1961. *Silence; Lectures and Writings*. Wesleyan University Press, Middletown, Conn.
- CLANCHY, M. T. 1979. *From Memory to Written Record*. Edward Arnold, London.
- CLARK, W. J. 1991. Spreadsheets versus data base management systems. *Fin. Acc. Syst.* (Spring), 10–14.
- COLLINS, H. M. 1990a. *Artificial Experts: Social Knowledge and Intelligent Machines*. The MIT Press, Cambridge, Mass.
- COLLINS, H. M. 1987. Expert systems and the science of knowledge. In *The Social Construction of Technological Systems*, W. E. Bijker, T. P. Hughes, T. Pinch, Eds., MIT Press, Cambridge, Mass., 329–348.

- COLLINS, W. P. 1990b. Effects of computerization requirements on the Baha'i classification scheme in use at the Baha'i World Center. *Cat. Class. Q.* 11, 2, 69ff.
- CRANE, G. 1988. Redefining the book: Some preliminary problems. *Acad. Comput.* (Feb), 6ff.
- DAFT, R. L. AND LENGEL, R. H. 1984. Information richness: A new approach to manager information processing and organizational design. In *Research in Organizational Behavior*, B. Staw and L. L. Cummings, Eds. JAI Press, Greenwich, Conn.
- DAVIS, G. B., COLLINS, R. W., EIERMAN, M., AND NANCE, W. D. 1991. Conceptual model for research on knowledge work. *MISRC Working Paper 91-10*, The Carlson School of Management, Univ. of Minnesota, Minneapolis, Minn.
- EISENSTEIN, E. L. 1983. *The Printing Revolution in Early Modern Europe*. Cambridge University Press, Cambridge, Mass.
- EISENSTEIN, E. L. 1979. *The Printing Press as an Agent of Change*. Cambridge University Press, Cambridge, Mass.
- GORRY, G. A., CHANEY, R. J., LONG, K. B., AND TAUSK, C. M. Computer support for biomedical workgroups. In *Proceedings of the Conference on Computer Supported Collaborative Work*. ACM, New York, 39-51.
- HENDERSON, K. 1991. Flexible sketches and inflexible data bases: Visual communication, conscription devices, and boundary objects in design engineering. *Sci. Tech. Hum. Val.* 16, 4, 448-473.
- IVINS, W. M. 1953. *Prints and Visual Communication*. Harvard University Press, Cambridge, Mass.
- KLING, R. AND SCACCHI, W. 1979. Recurrent dilemmas of computer use in complex organizations. In *Proceedings of the AFIPS National Computer Conference*. Vol. 48. AFIPS, 107-115.
- LAQUEY, T. AND RYER, J. C. 1993. *The Internet Companion: A Beginner's Guide to Global Networking*. Addison-Wesley, Reading, Mass.
- LATOUR, B. 1986. Visualization and cognition: Thinking with eyes and hands. *Knowl. Soc.: Stud. Soc. Cult. Past Pres.* 6, 1-40.
- LAVE, J. AND WENGER, E. 1991. *Situated learning: Legitimate, peripheral, participation*. Cambridge University Press, Cambridge, Mass.
- LEONARD-BARTON, D. AND SVOIKLA, J. J. 1988. Putting expert systems to work. *Harvard Bus. Rev.* 66, 2, 91-98.
- LEVESON, N. G. AND TURNER, C. S. 1992. An investigation of the Therac-25 accidents. UCI Tech. Rep. 92-108 (Univ. of Washington TR 92-11-05), Dept. of Information and Computer Sciences, Univ. of California, Irvine, Calif.
- MCLUHAN, M. 1964. *Understanding Media: The Extensions of Man*. McGraw-Hill, New York.
- MONMONIER, M. 1991. *How to Lie with Maps*. University of Chicago Press, Chicago, Ill.
- ONG, W. J. 1982. *Orality and Literacy*. Methuen, London.
- PERROLLE, J. 1986. Intellectual assembly lines: The rationalization of managerial, technical, and professional work. *Comput. Soc. Sci.* 2, 3, 111-122.
- RAGIN, C. C. AND BECKER, H. 1989. How the microcomputer is changing our analytic habits. In *New Technology in Sociology: Practical Applications in Research and Work*, G. Blank, J. L. McCartney, E. Brent, Eds. Transaction Publishers, New Brunswick, N.J.
- REACH. 1991. *Res. Educ. Appl. Comput. Hum.* Certificate in computing at University of Toronto. (Jan.-Feb.), 3.
- RENGSTORF, K. H. 1973. *A Complete Concordance to Flavius Josephus*. E. J. Brill, Leiden, Netherlands.
- REYNOLDS, L. D. AND WILSON, N. G. 1974. *Scribes and scholars: A guide to the transmission of Greek and Latin literature*. Clarendon Press, Oxford.
- RUHLER, K. 1995. The impact of computerization on knowledge production-A case study of classical scholarship. WPI-MG Working Paper 93-004, Dept. of Management, Worcester Polytechnic Inst., Worcester, Mass. In TDH204 Information Technology and Society CD-ROM, Open University, Milton Keynes, UK. Forthcoming.
- RUHLER, K. 1991. Information technology as instrument of social transformation. Ph.D dissertation, Univ. of California, Irvine, Calif.
- SCHATZ, B. 1991. Building an electronic community system. *J. Manage. Inf. Syst.* 8, 1, 87-107.

- STAR, S. L. 1990. Layered space, formal representations and long-distance control: The politics of information. *Fundamenta Scientia* 10, 2.
- SVIOKLA, J. J. 1990. An examination of the impact of expert systems on the firm: The case of XCON. *MIS Q.* 14, 2, 127-139.
- TRAWEEK, S. 1988. *Beamtimes and Lifetimes*. Harvard University Press, Cambridge, Mass.
- VON WILAMOWITZ-MOELLENDORFF, U. 1982. *History of Classical Scholarship*. Duckworth, London.
- WILKES, M. 1988. *The Concept of Disciple in Matthew's Gospel as Reflected in the Use of the Term [Mathetes]*. E. J. Brill, Leiden, Netherlands.
- WILL, R. 1991. True and false dependence on technology: Evaluation with an expert system. *Comput. Hum. Behav.* 7, 3, 171-183.
- YATES, J. AND ORLIKOWSKI, W. J. 1992. Genres of organizational communication: A structural approach to studying communication and media. *Acad. Manage. Rev.* 17, 2, 299-326.
- YOUTIE, H. C. 1974. *The Textual Criticism of Documentary Papyri* 2nd ed. Institute of Classical Studies, London.
- ZUBOFF, S. 1988. *In the Age of the Smart Machine: The Future of Work and Power*. Basic Books, New York.

Received December 1992; revised September 1993; accepted December 1993